IC4484 FAULT DETECTOR

The fault detector is designed to shut down an SCR control on two types of faults as follows:

(1) 1A CONTACTOR POWER TIPS FAIL TO OPEN

When the truck is in 1A (top speed) the 1A switchette in the accelerator is open, giving the fault detector card no signal to terminal 5. When the accelerator is brought back into the SCR range the 1A switchette closes. If 1A fails to open, battery negative is applied to point 5 of the card through T2 - T1 and if the fault does not clear within approximately 0.1 second, the circuit breaker trips, removing power by dropping out F or R contactor.

(2) SCR FAILURE WHICH CAUSES FULL MOTOR TORQUE

The SCR fails to pulse; it turns on and stays on continuously, causing essentially battery negative to be applied to point 5 of the card, similar to (1) described above. If such an SCR failure occurs the fault detector trips in 0.1 second.

TESTING

Jack the drive wheels up and operate the control in SCR range. Using an insulated tool, manually close 1A contactor and the fault detector should trip. If the fault detector fails to operate, check the following:

- Check 1A switchette in accelerator for proper operation.
- Remove the guard from the circuit breaker knob. Disconnect the wires from card points 1 and 2. Connect an ohmmeter to the circuit breaker contact points and manually operate the circuit breaker knob. The set position should read normally closed and the trip position should read normally open.

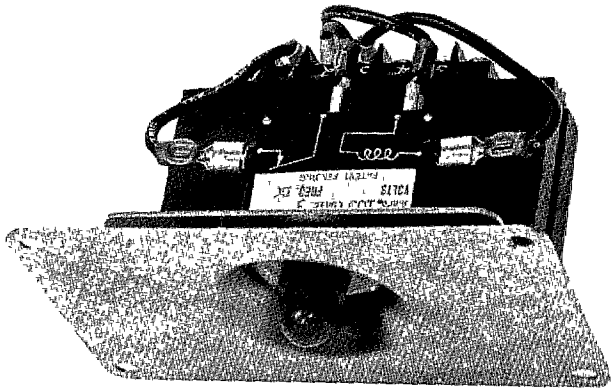


Fig. 1. IC4484D fault detector

- Disconnect the wires from card points 3 and 4. Measure circuit breaker coil resistance, approximately 60 ohms.

If all tests are good and the fault detector still fails to operate, replace the card.

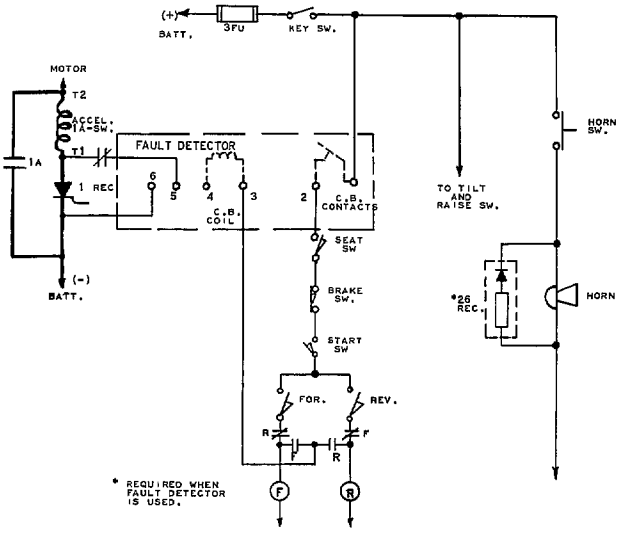


Fig. 2. Typical circuit showing fault detector.

The information contained herein is intended to assist truck users and dealers in the servicing of control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment or provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

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